

Morgan Kim

+1 (555) 010-8173 | morgan.kim@example.com | example.com/code/mkim

EDUCATION

Clearwater Institute of Technology | B.S. in Electrical Engineering | Metro City, USA | Sep 2017 - May 2021

EXPERIENCE

Northside Bakery | Delivery Driver | Metro City, USA | Aug 2025 - Present

- Delivered morning wholesale orders to about 20 caf s and collected signed invoices.
- Loaded the van and checked each order against the dispatch list before leaving.

Clearwater Sensors | Hardware Test Engineer | Metro City, USA | Jul 2021 - Feb 2023

- Made the sensor line's UART link to the test PC more reliable by raising its baud rate from 115200 to 921600.
- Traced a batch of early field failures to a solder-paste change; my findings updated the reflow profile and cut returns from 2.1% to 0.4%.
- Wrote the calibration procedure for the sensor line, cutting calibration drift complaints from 15 to 2 a quarter.
- Designed a test fixture that checked 8 boards at once while also ordering lab parts, maintaining the ESD program, training line operators and running the monthly safety walk, which raised station throughput from 40 to 150 boards an hour.

Voltline Robotics | Embedded Software Engineer | Metro City, USA | Mar 2023 - Jun 2025

- Rewrote the motor-controller firmware for a warehouse robot fleet of 300 units, cutting control-loop jitter from 120 to 15 microseconds and field motor faults by 60%.
- Cut firmware update time over the air from 11 minutes to 4 by compressing images and resuming interrupted downloads.
- Debounced the robot's emergency-stop input with a single 10 kOhm pull-up resistor.
- Profiling the interrupt load, moving sensor reads to DMA, splitting the control loop into fixed-priority tasks and pinning the scheduler tick, raised the controller's spare CPU from 8% to 35%.
- Presented the firmware roadmap to the hardware and product teams each quarter, and two board revisions followed its recommendations.
- Documented the bootloader and flashing process, so new engineers could flash a board on their first day.

PROJECTS

Low-Power Weather Station | Independent Project | C, STM32 | Jan 2025 - Present

- Built a solar-powered weather station on an STM32 that reports every 5 minutes over LoRa and runs 9 days without sun.
- Cut average current from 4 mA to 1 mA, a 300% reduction, by sleeping between readings and batching radio transmissions.
- Raised packet delivery by 15% over LoRa, from 80% to 95% of readings, by retrying failed transmissions.

Formula Student Electric Car | Electronics Lead, Team of 12 | C | Sep 2019 - May 2021

- Designed the battery-monitoring board for the team's electric car, which passed technical inspection on the first attempt.
- Wrote the CAN-bus firmware for the dashboard and tested it against the motor controller before each race.
- Mentored 4 first-year members on soldering and board bring-up.

SKILLS

Tools: C, C++, STM32, oscilloscopes

Methods: firmware, hardware-in-the-loop testing, CAN, low-power design